MILITARY SPECIFICATION

FUZE, TIME, XM768
METAL PARTS
LOADING ASSEMBLING AND PACKING

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions of the metal parts and loading, assembling and packing for one type of time fuze designated as Fuze, Time, XM768, formerly designated the M84A1E1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-12951	- Primer, Percussion, M39A1 Parts for, and Loading, Assembly and Packing
MIL-A-48078	- Ammunition, Standard Quality Assurance-Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-453	- Inspection, Radiographic
MIL-STD-1234	- Pyrotechnics, Sampling, Inspection and Testing
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

8865235	- Container, Ammunition, Fiber, M200A1 for M84A1 Time Fuze
8865236	- Box, Packing, Ammunition, for M84A1 Time Fuze in Fiber Container, M200A1
9349500	- Fuze, Time, XM768, Loading Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.1.1 Black powder. The moisture content of the black powder shall not exceed 0.7 percent.

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3.1.2 Pellet. The volatile content of the pellet shall not exceed 0.7 percent.

3.1.3 Tungsten delay composition. The volatile content of the delay composition shall not exceed 0.1 percent.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing (dwg.) 9349500 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein. (See 4.3 and 6.2).

3.4 Shear pin load. The shear pin when tested in accordance with paragraph 4.5.10 (dwg. 9349511) shall not shear when subjected to a minimum (min.) load of 25 pounds and shall shear when subjected to a maximum (max.) load of 41 pounds.

3.5 Die castings. The die castings shall be of uniform quality, free of cold shuts, blowholes and shrinkage defects. The casting must be dry, clean, and free from chips. All unfilled castings shall be rejected. The contractor, as part of his in-process control for die cast material, shall incorporate die cast procedures into his quality control manual.

3.5.1 Flash removal. All flash must be completely removed from parting lines, internal and external profiles, cored holes and ejector pin marks, except as noted on drawings.

3.5.2 Gates. All waste material remaining from the gate must be removed except when so noted on the drawing.

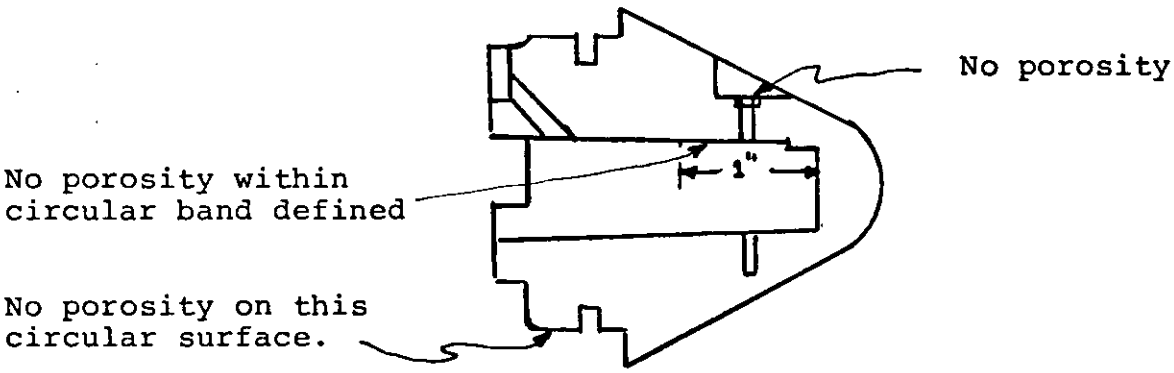
3.5.3 Surface porosity. Surface porosity is defined as metallic discontinuities called voids or pores. Individual pores shall be limited in number and in size to the requirements applicable to each component. A surface is defined as any area visible on the completed component. These surfaces include cavities, threads and surfaces revealed after completion of all machining operations:

	Government
Body, dwg.	9349519
Delay ring, dwg.	9349534
Head, dwg.	9349503
Primer Holder, dwg.	9349517

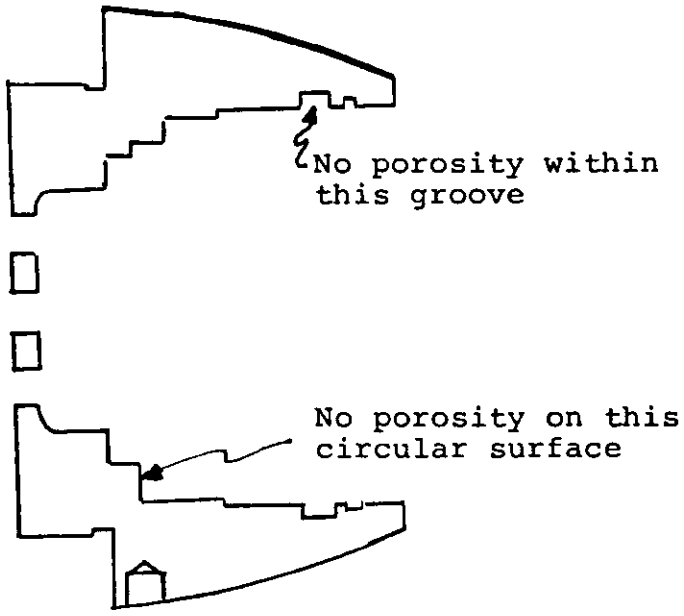
One pore shall be permitted no greater than .020 inch by .050 inch by .020 inch deep in any area measuring 1/4 inch by 1/4 inch except in the following areas where no surface porosity is permitted on the indicated surfaces of the three configurations shown.

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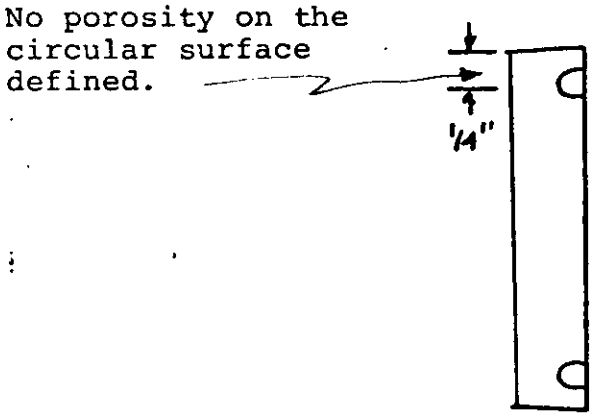
3.5.3.1 Head.



3.5.3.2 Body.



3.5.3.3 Delay Ring.



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3.6 Jolt and jumble test. The loaded fuze assembly shall comply with the requirements of Test 101.1 and 102.1 of MIL-STD-331.

3.7 Static functioning test. The loaded fuze assembly shall function and meet the burning time parameters of paragraph 4.4.3.6 when tested in accordance with paragraph 4.5.6.

3.8 Ballistic performance (Proving Ground). The loaded fuze assembly shall meet the following requirements.

3.8.1 Dud. The loaded fuze assembly shall not fail to function as intended.

3.8.2 Burning time. The loaded fuze assembly shall meet the burning time parameters of paragraph 4.4.3.7 when tested in accordance with paragraph 4.5.7.

3.8.3 Premature. The loaded fuze assembly shall not function nor shall the metal parts fail in the mortar or within 200 yards, horizontal range, of the mortar.

3.9 Leak test. The loaded fuze assembly shall have a leak rate of not greater than 0.5 pounds per square inch (psi) in 15 seconds minimum after pressure stabilization using a fixed volume of air of 3 psi $\pm$ 0.5 psi when tested in accordance with 4.5.9.

3.10 Torque test. The loaded fuze assembly shall be capable of being set between 6.5 to 41.5 inch-pounds at 70°F.

3.11 Radiographic inspection. The loaded fuze assemblies shall contain a complete ring delay assembly.

3.12 Workmanship. Parts and assemblies shall be free from chips, dirt, grease and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detailed specification.

4.1.1 Responsiblity for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all

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requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either acceptance of defective material.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item.

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings Government	Quantity
Disk	9349535	10
Body Assembly	9349518	10
Fuze, Time, XM768	9349500	322
Ring, Delay Assembly	9349526	10
Seal, Ring	9349522	10
Wrench, Fuze Setting	9349539	10
Ring, Spacer	9349521	10
Body	9349519	10/Cavity*
Head Subassembly	9349502	10
Primer Holder Assembly	9349516	10
Pellet, Ignitor	9349515	10
Pellet (Delay Ring)	9349528	10
Cover Assembly	9349513	10
Cover	9349514	10
Ring, Delay	9349534	10/Cavity*
Cup	9349525	10
Lid	9349524	10
Ring, Snap	9349512	10
Head	9349503	10/Cavity*
Firing Mechanism Assembly	9349507	200
Pull Wire & Seal Assembly	9349504	10
Holder Primer	9349517	10/Cavity*
Head Assembly Loading	9349501	10
Expulsion Charge Assembly	9349523	10
Pin, Shear	9349511	10
Guide, Plunger	9349508	10
Plunger	9349509	10
Clip, Lockout	9349510	10
Washer	9349506	10
Wire, Pull	9349505	10

Additionally, the contractor shall submit a certification that the following components have been checked and have been found acceptable by him.

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MS-9068-137	"O" Ring
9349538	Lubricant
9349520	Tape
MS-35671-B	Pin
9349537	Retainer
87989195	M39 Primer
MS-9068-13	"O" Ring
MS-9021-003	"O" Ring

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein. In addition, the provisions of paragraph 4.4.3.2.1 shall apply to characteristics identified by the contractor as die cast controlled inspections for the components marked with an asterisk (*) above.

4.3.3 Rejection. See MIL-A-48078.

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Table I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Fuze, Time, XM768		SHEET 1 OF 6		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Disc, Paper Dwg. 9349535 Examination for defects	10		3.2	4.4.2.2
	Body, Assembly Dwg. 9349518 Examination for defects	10		3.2	4.4.2.1
	Expulsion Charge Ass'y Dwg. 9349523 Examination for defects	10		3.2	4.4.2.8
	Seal Ring Dwg. 9349522 Examination for defects	10		3.2	4.4.2.9
	Wrench, Fuze Setting Dwg. 9349539 Examination for defects	10		3.2	4.4.2.10
	Ring, Spacer Dwg. 9349521 Examination for defects	10		3.2	4.4.2.11

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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Table I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Fuze, Time, XM768 (Cont)		SHEET 2 OF 6		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body Dwg. 9349519 Examination for defects	10/ cavity		3.2	4.4.2.12
	Porosity			3.5.3	4.5.11
	Head Sub-Assembly Metal Parts Dwg. 9349502 Examination for defects	10		3.2	4.4.2.13
	Cover Dwg. 9349514 Examination for defects	10		3.2	4.4.2.18
	Ring, Delay Dwg. 9349534 Examination for defects	10/ cavity		3.2	4.4.2.19
	Porosity			3.5.3	4.5.11
	Primer Holder Assembly Dwg. 9349516 Examination for defects	10		3.2	4.4.2.14

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Fuze, Time, XM768 (Cont)		SHEET 3 OF 6		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cover Assembly Dwg. 9349513 Examination for defects	10		3.2	4.4.2.15
	Pellet, Delay Ring Dwg. 9349528 Examination for defects	10		3.2	4.4.2.16
	Pellet, Igniter Dwg. 9349515 Examination for defects	10		3.2	4.4.2.17
	Ring, Snap Dwg. 9349512 Examination for defects	10		3.2	4.4.2.21
	Head Dwg. 9349503 Examination for defects Porosity	10/ cavity		3.2 3.5.3	4.4.2.22 4.5.11
	Firing Mechanism Assembly Dwg. 9349507 Examination for defects Shear pin load test	10 200	 1/	3.2 3.4	4.4.2.23 4.5.10

NOTES: 1/ One or more defects for min. load and three or more defects for max load shall cause the sample to be rejected.

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Table I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Fuze, Time, XM768 (Cont)		SHEET 4 OF 6		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Holder, Primer Dwg. 9349517 Examination for defects Porosity	10/ cavity		3.2 3.5.3	4.4.2.25 4.5.11
	Guide, Plunger Dwg. 9349508 Examination for defects	10		3.2	4.4.2.27
	Cup Dwg. 9349525 Examination for defects	10		3.2	4.4.2.20
	Pull Wire and Seal Assembly Dwg. 9349504 Examination for defects	10		3.2	4.4.2.24
	Pin, Shear Dwg. 9349511 Examination for defects	10		3.2	4.4.2.26
	Clip, Lock Out Dwg. 9349510 Examination for defects	10		3.2	4.2.2.29

NOTES:

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Table I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 5 OF 6		DRAWING NUMBER
	Fuze, Time, XM768 (Cont)				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Washer Dwg. 9349506 Examinaton for defects	10		3.2	4.4.2.30
	Wire, Pull Dwg. 9349505 Examination for defects	10		3.2	4.4.2.31
	Lid Dwg. 9349524 Examination for defects	10		3.2	4.4.2.32
	Plunger Dwg. 9349509 Examination for defects	10		3.2	4.4.2.28
	Delay Ring Assembly Dwg. 9349526 Examination for defects	10		3.2	4.4.2.7
	Head Assembly Loading Dwg. 9349501 Examination for defects	10		3.2	4.4.2.6

NOTES:

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Table I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Fuze, Time, XM768 (Cont)		SHEET 6 OF 6		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fuze, Time, XM768 Dwg. 9349500 Examination for defects	12 1/		3.2	4.4.2.5
	Jolt and Jumble Test Dwg. 9349500 Examination for defects	12		3.6	4.4.2.5 2/
	Static Functioning Test Dwg. 9349500 Observation for functioning Observation for burning time	200		3.7	4.5.6
	Ballistic Test Dwg. 9349500 Observation for prematures Observation for functioning Observation for burning time	100		3.8	4.5.7
	Leak Test Dwg. 9349500 Observation for leak	10		3.9	4.5.9

NOTES:
1/ Same samples used for jolt and jumble test.
2/ Conduct examination for Special C, paragraph 4.4.2.5.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of fuzes shall contain:

a. Primers of one lot interfix number from one manufacturer.

b. Each of the Ingredients used in the explosive and deflagrating components shall be from one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Die cast parts. If requested by the contractor and approved by the contracting officer, sampling for major and minor defects for die cast parts may be performed on the defects annotated by 1/ in the classification of defects paragraphs after successful completion of first article and the first production lot. See production verification of mold or die. If any parts are found to be defective, the cavity producing the defective part shall be removed from production. Also, that portion of production since the last successful quality conformance inspection shall be returned to the contractor for corrective action.

Cavities removed from production because of a fault, may, after reworking, be returned to production after passing the cast part die qualification inspection.

The contracting officer reserves the right to require the return to inspection by MIL-STD-105 sampling plans at any time and at no increase in obligation by the Government.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Body Assembly				9349518
					NEXT HIGHER ASSEMBLY 9349500
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Delay ring pin improperly assembled to body or bent or loose		0.40%	3.2	Visual/Manual
102	Height of pin		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Disk, Paper		9349535
			NEXT HIGHER ASSEMBLY 9349526

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter of disk	25	1/	3.2	Gage
202	Evidence of poor workmanship	25	1/	3.12	Visual

NOTES: 1/ Two or more defects shall be cause for lot rejection.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Fuze, Time XM768 (Prior to Assembly of Delay Ring and Head Assembly)				9349500
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Expulsion charge assembly missing		0.40%	3.2	Visual
102	Spacer ring missing		0.40%	3.2	Visual
103	Ring seal missing or assembled improperly		0.40%	3.2	Visual
104	O-ring lubricant missing		0.40%	3.2	Visual
105	Location pin missing or below height of seal ring		0.40%	3.2	Visual
106	Environmental conditions inadequate	1/		3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES: 1/ Temperature and Humidity shall be checked at the beginning and end of each shift. If conditions do not meet 40% maximum relative humidity at 72°F at either time, the quantity of product represented by the conditioning shall be rejected for further processing.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-70601 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.4	Fuze, Time XM768 (Prior to Assembly of Head Assembly)		1 OF 1	9349500
				NEXT HIGHER ASSEMBLY
				>

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Delay ring assembly missing or not properly engaging the location pin	1/	100%	3.2	Visual/Gage
Major					
101	Environmental conditions inadequate	2/	100%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

1/ The delay ring shall be gaged by a fail safe probe (designed to fail so as to indicate a defect). The body assembly shall be constrained in a mechanical system from the time of probe acceptance inspection until the head assembly is locked in place. Removal of the delay ring shall not be possible without reprobing. Effectiveness shall be checked at the beginning and end of each shift by insertion of easily recognized defective inert units. During probing each acceptable unit shall be color marked.

2/ Temperature and humidity shall be checked at the beginning and end of each shift. If conditions do not meet 40% maximum relative humidity at 72°F at either time, the quantity of product represented by the conditioning shall be rejected for further processing.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Fuze, Time, XM768	1 OF 2	9349500
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Delay ring missing	4/	100%	3.11	Radiographic
Special					
A	Fuze not set in safe position		100%	3.2	Visual
B	Pull wire assembly missing or improperly engaged		100%	3.2	Visual
C	Jolt and jumble test				
	Examination for defects	12	2/	3.6	4.5.2
D	Color mark missing from probe inspection		100%	3.2	Visual
Major					
101	Tape missing, not completely covering holes or improperly affixed		100%	3.2	Visual
102	Leak Test				
	Observation for leak	10	1/	3.9	4.5.9
103	Head improperly assembled to body (evidenced by excessive gap between head and body; and snap ring ends incorrectly positioned		0.40%	3.2	Visual
104	Environmental conditions inadequate	3/		3.2	Gage
105	Head torque (observed during 360° rotations)		100%	3.10	Torque wrench

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.5	Fuze, Time, XM768 (Cont)				9349500
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Marking missing or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.12	Visual

1/ The lot shall be rejected if one or more defectives are found. 2/ The lot shall be rejected if one or more fuzes function or are found to be in an unsafe condition. 3/ Temperature and humidity shall be checked at the beginning and end of each shift. If conditions do not meet drawing requirements at either time, the quantity of product represented by the sample shall be rejected for further processing. 4/ This inspection may be deleted provided the probe (see para 4.4.2.4, Crit 1) inspection is proven effective to the satisfaction of the government inspection equipment office (see para 6.3).

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349501
4.4.2.6	Head Assembly Loading				NEXT HIGHER ASSEMBLY 9349500
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cover assembly missing, loose or pellet damaged		0.40%	3.2	Visual
102	Primer holder assembly missing		0.40%	3.2	Visual
103	Snap ring missing		0.40%	3.2	Visual
104	Primer holder assembly not flush or below flush		0.40%	3.2	Visual
105	Environmental conditions inadequate	1/		3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES: 1/ Temperature and humidity shall be checked at the beginning and end of each shift. If conditions do not meet drawing requirements at either time, the quantity of product represented by the sample shall be rejected for further processing.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Delay Ring Assembly		9349526
			NEXT HIGHER ASSEMBLY 9349500

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter over max.		0.40%	3.2	Gage
102	Delay composition above flush		0.40%	3.2	Visual
103	Component parts missing or loose or all spokes in retaining ring not engaged		0.40%	3.2	Visual/Manual
104	Thickness of delay ring		0.40%	3.2	Gage
105	Consolidation Pressure (delay comp)	1/		3.2	Gage
106	Evidence of cracks, pits, or voids in tungsten composition		0.15%	3.2	Visual
107	Facing machine cutting marks across tungston delay		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES: 1/ Actual Pressure and tolerances to be defined by contractor. Inspection for pressure to be performed twice a shift. All assemblies since last acceptable inspection shall be rejected if pressure exceeds limits.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349523
4.4.2.8	Expulsion Charge Assembly				NEXT HIGHER ASSEMBLY 9349500
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of lid diameter to cup diameter		0.40%	3.2	Gage
102	Charge weight under min.	20	1/	3.2	Balance
103	Cup and lid improperly sealed or mated	2/	0.40%	3.2	Visual/Manual
104	Absence of black powder in assembly		100%	3.2	Visual
105	Tears, holes, or rips in assembly		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES: 1/ One or more defects will cause lot rejection. 2/ Seal can be checked by applying a light pressure on top and bottom of cup using thumb and index finger and observing for air leakage.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.9	Seal, Ring	1	1	9349522
				NEXT HIGHER ASSEMBLY: 9349500

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter of ring		0.40%	3.2	Gage
102	Thickness of ring		0.40%	3.2	Gage
103	Width of ring		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Wrench, Fuze Setting		9349539
			NEXT HIGHER ASSEMBLY 9349500

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width of wrench slot		0.40%	3.2	Gage
102	Depth of wrench slot		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349521
4.4.2.11	Ring, Spacer				NEXT HIGHER ASSEMBLY 9349500
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness, minimum	1/	0.40%	3.2	Gage
Minor					
201	Outside diameter	1/	0.65%	3.2	Gage
202	Inside diameter	1/	0.65%	3.2	Gage
203	Evidence of poor workmanship	1/	0.65%	3.12	Visual

NOTES: 1/ Sample size shall be selected in accordance with Special Inspection Level S-1, Table II-A of MIL-STD-105.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-70601 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9349519
4.4.2.12	Body				NEXT HIGHER ASSEMBLY 9349500
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Location of set mark relative to delay ring locating lug	1/	100%	3.2	Gage 2/
Major					
101	Diameter of snap ring slot	1/	0.40%	3.2	Gage
102	Largest cavity inside diameter	1/	0.40%	3.2	Gage
103	Diameter of o-ring slot	1/	0.40%	3.2	Gage
104	Distance from front face to snap ring slot	1/	0.40%	3.2	Gage
105	Width of snap ring slot	1/	0.40%	3.2	Gage
106	Width of o-ring slot	1/	0.40%	3.2	Gage
107	Distance from front face to head cavity shoulder, min.	1/	0.40%	3.2	Gage
108	Distance from front face of snap ring slot to shoulder of delay ring assembly cavity	1/	0.40%	3.2	Gage
109	Length of delay ring locating lug	1/	0.40%	3.2	Gage
110	Depth of delay ring locating lug	1/	0.40%	3.2	Gage
111	Diameter of expulsion charge assembly cavity	1/	0.40%	3.2	Gage
112	Perpendicularity of outside shoulder forward of thread to pitch diameter of thread	1/	0.40%	3.2	Gage

NOTES: 1/ If inspection is based on product verification of mold or die, see 4.4.3.2.2 for sample size and reject criteria. 2/ All rejects shall be held for Gov't disposition.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9349519
4.4.2.12	Body (Cont)				NEXT HIGHER ASSEMBLY 9349500
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Major diameter of threads min.	1/	0.40%	3.2	Gage
114	Minor diameter of threads max	1/	0.40%	3.2	Gage
115	Pitch diameter of threads	1/	0.40%	3.2	Gage
116	Location of pin hole	1/	0.40%	3.2	Gage
117	Diameter of pin hole	1/	0.40%	3.2	Gage
118	Depth of pin hole	1/	0.40%	3.2	Gage
119	Width of delay ring locating lug	1/	0.40%	3.2	Gage
120	Angle between centerline of delay ring locating lug and pin hole center	1/	0.40%	3.2	Gage
Minor					
201	Depth of spacer ring cavity	1/	0.65%	3.2	Gage
202	Distance from delay ring assembly cavity shoulder to front edge of o-ring slot	1/	0.65%	3.2	Gage
203	Length of body	1/	0.65%	3.2	Gage
204	Outer diameter of body	1/	0.65%	3.2	Gage
205	Diameter of ring spacer cavity, min.	1/	0.65%	3.2	Gage
206	Diameter of delay ring assembly cavity, min. (forward and aft)	1/	0.65%	3.2	Gage
207	Flash hole missing or improperly located	1/	0.65%	3.2	Visual
208	Evidence of poor workmanship	1/	0.65%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-70601 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Head Sub-Assembly Metal Parts				9349502
					NEXT HIGHER ASSEMBLY 9349500
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pull wire and seal assembly missing or not properly engaged		0.40%	3.2	Visual
102	O-ring missing or not properly lubricated		0.40%	3.2	Visual
103	Snap ring missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349516
4.4.2.14	Primer Holder Assembly				NEXT HIGHER ASSEMBLY 9349501
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
A	Percussion primer missing or loose or not fully seated	1/ 2/	100%	3.2	Visual/Manual
Major					
101	Environmental conditions inadequate	3/		3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.12	Visual

1/ Primer holder assembly shall be held with primer in down position and the assembly shall be shook 2 or 3 times. The primer shall not move in its seat. 2/ All proper safety precautions must be exercised to preclude inadvertant primer firing. 3/ The temperature and humidity shall be checked at the beginning and end of each shift. If conditions do not meet drawing requirements at either time, the quantity of product represented by the sample shall be rejected for further processing.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.15	Cover Assembly		9349513
			NEXT HIGHER ASSEMBLY 9349501

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pellet cracked, loose, or missing		0.40%	3.2	Visual
102	Pellet above flush		0.40%	3.2	Visual
103	Pellet weight	10	1/	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES: 1/ If one or more of the pellets do not meet the drawing weight the quantity of product represented by the sample shall be rejected for further processing.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Pellet (Delay Ring)				9349528
					NEXT HIGHER ASSEMBLY 9349526
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of pellet		0.40%	3.2	Gage
102	Thickness of pellet		0.40%	3.2	Gage
103	Density of pellet under min.	10	1/	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES: 1/ One or more defects shall be cause for rejection of the quantity represented by the sample.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.17	Pellet, Ignitor (If Applicable)		9349515
			NEXT HIGHER ASSEMBLY 9349514

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width of pellet		0.40%	3.2	Gage
102	Length of pellet		0.40%	3.2	Gage
103	Incorrect pellet radii		0.40%	3.2	Gage
104	Pellet cracked or laminated		0.40%	3.2	Visual
105	Pellet thickness		0.40%	3.2	Gage
106	Pellet consolidation pressure incorrect	1/		3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

1/ Pressure shall be checked twice each shift, once at the beginning and once at the end. Material shall not be consolidated until the pressure is correct or all material shall be rejected since the last proper pressure was recorded if the pressure is incorrect at the end of the shift.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349514
4.4.2.18	Cover				NEXT HIGHER ASSEMBLY 9349500
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large outside radius	1/	0.40%	3.2	Gage
102	Small outside radius	1/	0.40%	3.2	Gage
103	Total length	1/	0.40%	3.2	Gage
104	Outside width	1/	0.40%	3.2	Gage
105	Length between longitudinal edges at large outside radius	1/	0.40%	3.2	Gage
106	Radius of pellet cavity slot	1/	0.40%	3.2	Gage
107	Width of thru cavity slot	1/	0.40%	3.2	Gage
108	Location of thru cavity slot	1/	0.40%	3.2	Gage
109	Length from pellet end of cover to ear flaps	1/	0.40%	3.2	Gage
110	Height of cover flaps	1/	0.40%	3.2	Gage
111	Radius of index tongue	1/	0.40%	3.2	Gage
112	Distance from rear face to pellet cavity face	1/	0.40%	3.2	Gage
Minor					
201	Location of first step of pellet cavity	1/	0.65%	3.2	Gage
202	Location of depression around pellet cavity	1/	0.65%	3.2	Gage
203	Radius of depression around pellet cavity	1/	0.65%	3.2	Gage
204	Evidence of poor workmanship	1/	0.65%	3.12	Visual

NOTES: 1/ Sample size shall be selected in accordance with Special Inspection Level S-1 Table II-A of MIL-STD-105.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.19	Ring, Delay		9349534
			NEXT HIGHER ASSEMBLY 9349526

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
A	Angle between centerline of delay composition and end of delay composition channel	1/		3.2	Gage
B	Location of locating slot relative to centerline of delay composition channel	1/		3.2	Gage
Major					
101	Thickness	1/	0.40%	3.2	Gage
102	Profile of delay composition channel (including position relative to datum diameter)	1/	0.40%	3.2	Gage
103	Pellet hole missing or obstructed		0.40%	3.2	Visual
104	Rear outside diameter, maximum	1/	0.40%	3.2	Gage
105	Pellet hole inside diameter	1/	0.40%	3.2	Gage
106	Locating slot width, minimum.	1/	0.40%	3.2	Gage
107	Location slot depth, minimum	1/	0.40%	3.2	Gage
108	True position of outside rear diameter with forward outside diameter	1/	0.40%	3.2	Gage
109	Location of pin hole	1/	0.40%	3.2	Gage
110	Depth of pin hole	1/	0.40%	3.2	Gage
111	Diameter of pin hole	1/	0.40%	3.2	Gage

NOTES: 1/ If inspection is based on product verification of mold or die, see 4.4.3.2.2 for sample size and rejection criteria.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9349534
4.4.2.19	Ring, Delay (Cont)				NEXT HIGHER ASSEMBLY 9349526
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Diameter of delay composition channel	1/	0.40%	3.2	Gage
113	Width of delay composition channel	1/	0.40%	3.2	Gage
114	Depth of delay composition channel	1/	0.40%	3.2	Gage
115	Inner radius of channel adjacent to pellet hole	1/	0.40%	3.2	Gage
116	Outer radius of channel adjacent to pellet hole	1/	0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship	1/	0.65%	3.11	Visual
NOTES:					

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.20	Cup		9349525
			NEXT HIGHER ASSEMBLY 9349523

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall diameter		0.40%	3.2	Gage
102	Cup outside diameter		0.40%	3.2	Gage
103	Depth of cup		0.40%	3.2	Gage
104	Evidence of tears, rips, or perforations		0.40%	3.2	Visual
105	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.21	Ring, Snap		9349512
			NEXT HIGHER ASSEMBLY 9349502

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness, max.	1/	0.40%	3.2	Gage
102	Outside diameter, max.	1/	0.40%	3.2	Gage
103	Width of ring	1/	0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship	1/	0.65%	3.12	Visual

NOTES: 1/ Sample size shall be selected in accordance with Special Inspection Level S-1, Table II - A of MIL-STD-105.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER 9349503
4.4.2.22	Head		NEXT HIGHER ASSEMBLY 9349502

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
A	Presence of safe mark(s)		100%	3.2	Visual
B	Location of safe(s) mark relative to flash hole		100%	3.2	Gage
Major					
101	Second largest outside diameter, max.	1/	0.40%	3.2	Gage
102	Height of positioning lug	1/	0.40%	3.2	Gage
103	Length from base to bottom of firing mechanism assembly positioning lug, max.	1/	0.40%	3.2	Gage
104	Length from base to snap ring groove, max.	1/	0.40%	3.2	Gage
105	Radial location of timing marks relative to flash hole and accumulated error	1/	0.40%	3.2	Gage
106	Flash hole missing or obstructed	1/	0.40%	3.2	Visual
107	Plunger guide locating lug width, max.	1/	0.40%	3.2	Gage
108	Width of snap ring groove, min.	1/	0.40%	3.2	Gage
109	True position of plunger guide locating lug	1/	0.40%	3.2	Gage
110	Largest inside diameter, min.	1/	0.40%	3.2	Gage
111	True position of plunger cavity	1/	0.40%	3.2	Gage
112	Diameter of firing mechanism assembly cavity at primer holder positioning lug, min. and location of lug	1/	0.40%	3.2	Gage

NOTES: 1/ If inspection is based on product verification of mold or die, see 4.4.3.2.2 for sampling size and reject criteria.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER 9349503
4.4.2.22	Head, (Cont)		NEXT HIGHER ASSEMBLY 9349502

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Depth of safety wire cavity	1/	0.40%	3.2	Gage
114	Inner daimeter of snap ring groove	1/	0.40%	3.2	Gage
115	True position of snap ring groove	1/	0.40%	3.2	Gage
116	Diameter of safety wire cavity o-ring notch, min.	1/	0.40%	3.2	Gage
117	Diameter of safety wire cavity	1/	0.40%	3.2	Gage
118	Depth of flash hole groove	1/	0.40%	3.2	Gage
119	Width of flash hole groove	1/	0.40%	3.2	Gage
120	Location of flash hole groove	1/	0.40%	3.2	Gage
121	Radius of cover holder	1/	0.40%	3.2	Gage
122	Distance from body center axis to inner face of flash hole groove holder	1/	0.40%	3.2	Gage
123	Length of flash hole groove	1/	0.40%	3.2	Gage
124	Diameter of flash hole	1/	0.40%	3.2	Gage
125	Distance from center axis of flash hole to outside edge of flash hole groove holder at outer edge of body	1/	0.40%	3.2	Gage
126	Width of flash hole groove holder at outer edge of body	1/	0.40%	3.2	Gage
127	Set markings illegible or improperly applied	1/	0.40%	3.2	Visual
Minor 201	Distance from front edge of body to seating shoulder	1/	0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9349503
4.4.2.22	Head, (Cont)				NEXT HIGHER ASSEMBLY 9349502
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
202	Safety wire cavity missing or obstructed	1/	0.65%	3.2	Visual
203	Overall length of body	1/	0.65%	3.2	Gage
204	Width of flash hole groove holder at inside edge	1/	0.65%	3.2	Gage
205	Width of cover holder	1/	0.65%	3.2	Gage
206	Largest diameter of body, min.	1/	0.65%	3.2	Gage
207	Evidence of poor workmanship	1/	0.65%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349507
4.4.2.23	Firing Mechanism Assembly				NEXT HIGHER ASSEMBLY 9349502
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Shear pin missing or not staked or improperly staked		100%	3.2	Visual
2	Shear pin load test, minimum load	1/		3.4	4.4.3.1/4.5.10
Major					
101	O-ring missing		0.40%	3.2	Visual
102	Plunger missing or inverted		0.40%	3.2	Visual
103	Clip missing		0.40%	3.2	Visual
104	Shear pin load test, maximum load	1/		3.4	4.4.3.1/4.5.10
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES: 1/ See paragraph 4.4.3.1 for sample size and reject criteria. Same samples shall be used for both Critical 2 and Major 104 tests.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349504
4.4.2.24	Pull Wire and Seal Assembly				NEXT HIGHER ASSEMBLY 9349500
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	O-ring missing, loose, or improperly assembled		0.40%	3.2	Visual
102	Washer missing or improperly assembled		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.25	Holder, Primer	1	1	9349517
				NEXT HIGHER ASSEMBLY 9349516

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, max.	1/	0.40%	3.2	Gage
102	True position of flash hole	1/	0.40%	3.2	Gage
103	Outside diameter, forward	1/	0.40%	3.2	Gage
104	Outside diameter, aft	1/	0.40%	3.2	Gage
105	Distance from large flat to max taper of wedge	1/	0.40%	3.2	Gage
106	Diameter of primer hole	1/	0.40%	3.2	Gage
107	Depth of primer hole	1/	0.40%	3.2	Gage
108	Diameter of flash hole	1/	0.40%	3.2	Gage
109	Improper angle of flash hole	1/	0.40%	3.2	Gage
110	Concentricity of primer hole to small outside diameter	1/	0.40%	3.2	Gage
Minor					
201	Improper taper of flat	1/	0.65%	3.2	Gage
202	Length from flat to wedge	1/	0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES: 1/ If inspection is based on product verification of mold or die, see 4.4.3.2.2 for sampling size and reject criteria.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	Pin, Shear	1 OF 1	9349511
			NEXT HIGHER ASSEMBLY: 9349507

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of pin		0.40%	3.2	Gage
102	Diameter of pin		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.27	Guide, Plunger				9349508
					NEXT HIGHER ASSEMBLY 9349507
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, max.		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
103	O-ring groove diameter		0.40%	3.2	Gage
104	Width of o-ring groove		0.40%	3.2	Gage
105	True position of o-ring groove diameter to outside diameter		0.40%	3.2	Gage
106	Location of pull wire holes		0.40%	3.2	Visual
107	Diameter of pull wire holes		0.40%	3.2	Gage
108	Width of lock out clip groove		0.40%	3.2	Gage
109	Location of shear wire holes		0.40%	3.2	Gage
110	Diameter of shear wire holes		0.40%	3.2	Gage
111	Length of guide legs		0.40%	3.2	Gage
112	Inside diameter of guide legs		0.40%	3.2	Gage
113	Inside diameter of guide		0.40%	3.2	Gage
114	True position of locator slot		0.40%	3.2	Gage
Minor					
201	Improper inside diameter surface finish		0.65%	3.2	Gage
202	Width of locator slot		0.65%	3.2	Gage
203	Depth of locator slot		0.65%	3.2	Gage
204	Pull wire hole chamfers missing		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.28	Plunger	1 OF 1	9349509
			NEXT HIGHER ASSEMBLY 9349507

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height of firing pin		0.40%	3.2	Gage
102	Profile of firing pin		0.40%	3.2	Gage
103	Total length of plunger		0.40%	3.2	Gage
104	Outside diameter of plunger at firing pin face		0.40%	3.2	Gage
			0.40%	3.2	Gage
105	Location of pull wire holes		0.40%	3.2	Gage
106	Diameter of pull wire holes, min.		0.40%	3.2	Gage
107	Location of shear wire holes		0.40%	3.2	Gage
108	Diameter of shear wire holes		0.40%	3.2	Gage
109	Protective finish missing or incomplete		0.40%	3.2	Gage
Minor					
201	Distance between longitudinal flats		0.65%	3.2	Gage
202	Width of o-ring groove		0.65%	3.2	Gage
203	True position of o-ring groove flange		0.65%	3.2	Gage
204	Distance from firing pin face to o-ring groove		0.65%	3.2	Gage
			0.65%	3.2	Gage
205	Diameter of o-ring groove		0.65%	3.2	Visual
206	Chamfers missing or improper		0.65%	3.12	Visual
207	Evidence of poor workmanship				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.29	Clip, Lock Out				9349510
					NEXT HIGHER ASSEMBLY 9349507
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Protective finish missing or incomplete		0.65%	3.2	Visual
202	Outside width at widest point		0.65%	3.2	Gage
203	Length of clip		0.65%	3.2	Gage
204	Distance between legs		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.30	Washer				9349506
					NEXT HIGHER ASSEMBLY 9349504
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter of through hole		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349505
4.4.2.31	Wire, Pull				NEXT HIGHER ASSEMBLY 9349504
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of wire		0.40%	3.2	Gage
102	Evidence of embrittlement	20 1/		3.2	4.5.12
Minor					
201	Inside radius from center		0.65%	3.2	Gage
202	Length from washer stop to end of pin point		0.65%	3.2	Gage
203	Width of washer stop ears, min.		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

1/ One or more defects shall be cause for lot rejection.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.32	Lid		9349524
			NEXT HIGHER ASSEMBLY 9349523

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter of lid		0.65%	3.2	Gage
202	Thickness of lid		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables and as described below.

4.4.3.1 Shear pin load test.

4.4.3.1.1 Shear pin load test. Beginning with the first lot produced and continuing until three consecutive lots have been found to be acceptable, two hundred firing mechanism assemblies shall be randomly selected from each lot for testing. The lot shall be rejected if one or more assemblies fail the minimum load requirement or if three or more assemblies fail the maximum load requirement.

4.4.3.1.2 Shear pin load test. After three consecutive lots have complied with the criteria of 4.4.3.1.1, 125 firing mechanism assemblies shall be randomly selected from each lot for testing. The lot shall be rejected if one or more assemblies fail the minimum load requirement or if three or more assemblies fail the maximum load requirement. Lot failure shall cause the provisions of 4.4.3.1.1 to apply for subsequent lots.

4.4.3.2 Dimensional control of molded and die cast component.

4.4.3.2.1 Qualification of molds and dies. The examination and acceptance of the first article is generally accepted as qualification of the related molds and dies. Subsequent qualifications are mandatory in order to release molds and dies, or individual cavities which were rejected and removed from production due to producing a defective product, due to wear, breakage, etc. or because of a change in configuration of the component by a revision to the technical data package and the contract. A sample of ten consecutively produced components shall be selected from each mold and die cavity and examined in accordance with the applicable Classification of Defects paragraph as a minimum. If one or more components prove to be defective, the cavity producing the defective part shall not be released for production. Cavities not released for production and cavities removed from production because of forming defective components during a production run may, after repair or rework be subjected to the qualification test and, if found acceptable, be returned to production. A die containing multiple cavities and found to contain a defect producing cavity shall not be utilized in production until that cavity is repaired or a positive means of isolating that cavity from production is incorporated in the mold or die.

4.4.3.2.2 Production verification of mold or die. The first three consecutive produced components from each cavity produced during the same production run shall be examined in accordance with the applicable Classification of Defects identified by a

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1/ A production run is defined as the continuous forming of consecutive components by a mold or die which has been prepared in accordance with its operating specifications, and maintained in that condition from when the first sample is selected until the last sample (three parts) is formed which represents that production run. The production run represented by these two samples shall be rejected if either of the two samples are found to contain one or more defects. Samples shall be permitted to cool to room temperature, 80 ± 20 degrees Fahrenheit prior to examination for compliance to dimensional requirements.

A production run may be kept operating to produce all of the components required under the contract and the contractor may elect to cut-off certain quantities for deliveries. This is acceptable provided three components are selected at the end of that quantity. No production run shall be greater than one shift.

4.4.3.3 Black powder moisture content test. A two gram sample shall be selected from each lot of black powder used in each eight hours of fuze assembly production. A composite sample shall not be used. If the moisture content of the sample exceeds the requirement, the quantity of black powder represented by the sample shall be rejected. Also, if fuze assemblies have been loaded with the representative powder, the fuze assemblies shall be rejected. Testing shall be performed as specified in paragraph 4.5.3.

4.4.3.4 Volatile content of pellet. Prior to fabrication as a pellet, a two gram sample shall be selected from each lot of boron-potassium nitrate mixture comprising that quantity of pellet used in each of eight hours of fuze assembly production. A composite sample shall not be used. If the volatile content of the sample exceeds the requirement, the quantity of boron potassium nitrate represented by the sample shall be rejected. Testing shall be performed as specified in paragraph 4.5.4.

4.4.3.5 Volatile content of tungsten delay composition. Prior to loading the delay ring, a two gram sample shall be selected from each lot of tungsten delay composition mixture comprising that quantity of delay ring assemblies used in each eight hours of fuze assembly production. A composite sample shall not be used. If the volatile content of the sample exceeds the requirement, the quantity of tungsten delay composition represented by the sample shall be rejected. Testing shall be performed as specified in paragraph 4.5.5.

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4.4.3.6 Static functioning test.

4.4.3.6.1 Static functioning test. For First Article only, two hundred loaded fuze assemblies shall be tested in accordance with table II. The lot shall be rejected if two or more duds occur (total) or if the specified standard deviation (unbiased) within any group exceeds the values of table II. In addition the lot shall be rejected if the arithmetic average of the achieved burn time of any group exceeds plus or minus two percent of the applicable set time (engraved). Testing shall be performed as specified in paragraph 4.5.6.

Static Functioning Test

Table II

Temp.	-50°F		+70°F				+145°F	
Fuze set time (sec)	5	55	5	15	30	55	5	55
Quantity	25	25	25	25	25	25	25	25
Standard Deviation	.35	1.30	.27	.30	.58	.80	.24	.36

4.4.3.6.2 Static functioning test. Beginning with the first lot produced and continuing until three consecutive lots comply with the acceptance criteria, one hundred loaded fuze assemblies shall be randomly selected from each lot and tested in accordance with Table III. The lot shall be rejected if two or more duds occur (total) or if the specified standard deviation (unbiased) within any group exceeds the values of Table III. In addition the lot shall be rejected if the arithmetic average of the achieved burn time of any group exceeds plus or minus two percent of the applicable set time. Testing shall be performed as specified in paragraph 4.5.6.

Static Functioning Test

TABLE III

Fuze Set Time (sec)	5	15	30	55
Quantity	25	25	25	25
Standard Deviation	.27	.30	.58	.80

4.4.3.6.3 Static functioning test. After three consecutive lots comply with the criteria of 4.4.3.6.2, fifty loaded fuze assemblies shall be randomly selected from each lot. Twenty-five fuze assemblies shall be tested at the fifteen second setting and twenty-five at the thirty second setting. The lot shall be rejected if one or more duds occur (total) or if the standard deviation (unbiased) within each group exceeds the values of the appropriate setting of Table III. In addition the lot shall be rejected if the arithmetic average of the achieved burn time of any group exceeds plus or minus two percent of the applicable set time. Testing shall be performed as specified in paragraph 4.5.6.

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4.4.3.7 Proving ground test.

4.4.3.7.1 Proving ground test. Beginning with the first article, the first lot produced and continuing until three consecutive lots comply with the acceptance criteria, one hundred loaded fuze assemblies shall be randomly selected from each lot and tested to meet the requirements of Table III. The lot shall be rejected if three or more duds occur (total) or if the specified standard deviation (unbiased) within each group exceeds the values of Table III. In addition the lot shall be rejected if the arithmetic average of the achieved burn time of any group exceeds plus or minus two percent of the applicable set time. The lot shall also be rejected if any premature functionings occur. Testing shall be performed as specified in paragraph 4.5.7.

4.4.3.7.2 Proving ground test. After three consecutive lots have complied with the criteria of 4.4.3.7.1, fifty loaded fuze assemblies shall be randomly selected from each lot. Twenty-five fuze assemblies shall be tested at the fifteen second setting and twenty five at the thirty second setting. The lot shall be rejected if a total of two or more duds occur or if the specified standard deviation (unbiased) within each group exceeds the values of Table II. In addition the lot shall be rejected if the arithmetic average of the achieved burn time of any group exceeds plus or minus two percent of the applicable set time. The lot shall also be rejected if any premature functioning occurs. Testing shall be performed as specified in paragraph 4.5.7.

4.4.3.8 Check test for deterioration of percussion primer. If the total elapsed time between original acceptance of the percussion primer lot and the assembly of that lot into the fuze assembly exceeds two years, or if the percussion primers have been subjected to adverse conditions for any period of time since previous tests, the percussion primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-12951 (AR) immediately before the percussion primer lot is assembled into the fuze assembly.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1 and 4.4.3. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

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4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 and 4.4.3 herein and Table I.

4.5.2 Jolt and jumble test. The loaded fuze assembly shall be tested in accordance with test procedure 101.1 and 102.1 of MIL-STD-331. Examination to determine acceptability shall be conducted upon completion of both tests.

4.5.3 Black powder moisture content test. This test shall be performed in accordance with Method 102.2 of MIL-STD-1234.

4.5.4 Volatile content of pellet. This test shall be performed in accordance with Method 102.2 of MIL-STD-1234.

4.5.5 Volatile content of tungsten delay composition. This test shall be performed in accordance with Method 102.2 of MIL-STD-1234.

4.5.6 Static functioning test. The loaded fuze assembly shall be prepared by drilling a hole vertically through the head large enough to allow an appropriate initiation device to pass through to function the firing mechanism assembly. The fuze assembly shall then be conditioned at 70°F plus or minus 5°F for a minimum of four hours and tested immediately thereafter. For First Article testing the fuze shall be conditioned for a minimum of four hours at the specified temperature plus or minus 5°F and tested immediately thereafter. The fuze shall be functioned and fuze burning times shall be obtained by equipment capable of determining direct reading to the nearest .005 seconds. If any fuze assembly fails to comply with the requirements of 3.7, the assembly shall be classified defective. Data results from this test shall be submitted in accordance with DI-T-1906.

4.5.7 Ballistic test. The loaded fuze assembly shall be assembled to an 81MM XM819 or XM853 cartridge and fired from an 81MM M252 mortar as specified in Table IV. Cartridge/fuze combinations

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shall be conditioned a minimum of four hours at 70° plus or minus 5°F and fired immediately after removal from conditioning. Flight time to fuze functioning shall be recorded by an electronic method capable of accuracy to the nearest .05 seconds. Any cartridge/fuze combination which fails to meet the requirements of 3.8 and Table III shall be classified defective.

Ballistic Test

TABLE IV

Cartridge	Mortar	Zone	Set Time (sec)
XM853/XM819	M252	1	5
XM853/XM819	M252	2	15
XM853/XM819	M252	3	30
XM853/XM819	M252	4	55

Elevation shall be 60°

4.5.8 Delay ring assembly curing. Certified records that contain curing time, temperature and assembly lot number shall be maintained.

4.5.9 Leak test. The bare fuze shall be placed in an air pressure vessel that has an internal volume that is not greater than twice the total volume of the fuze. If fixtures are used in the vessel to hold the fuze, the fixture shall not impede the entrance of air into the fuze. The vessel shall be pressurized with air to 3 plus or minus .5 psi at ambient temperature and held for a minimum of 15 seconds. The pressure reading shall be observed. If the pressure fails to reach 3 plus or minus .5 psi or falls below 2.5 psi within 15 seconds, the sample shall be classified defective.

4.5.10 Shear pin load test. The firing mechanism assembly shall be inserted in a suitable holding fixture and placed in the testing machine. The required load shall be applied to the end of the plunger opposite the firing pin and shall be parallel to the major axis of the plunger. The load shall be applied at a travel rate of not less than 0.100 nor more than 0.125 inch per minute.

4.5.11 Porosity. Samples selected for porosity testing shall be visually inspected for surface defects.

4.5.12 Embrittlement test. The formed component shall be flexed to the point of minimum permanent deformation. The clip shall be flexed in a manner to straighten the component and the pull wire shall be flexed to bend the component until upon removal of the

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deforming pressure the clip no longer returns to its formed shape and the pull wire contains a permanent slight bend. Any component that breaks or splits prior to this condition shall be classed defective. Components subjected to this test shall not be returned to the lot. If one or more components fail the test, the batch of components shall be rejected.

4.5.13 Determination for presence of components. The sample shall be subjected to radiographic examination. The radiographic beam shall form a 90° angle with the longitudinal axis of the fuze. Examinations of the radiograph shall be made to determine presence of the delay ring assy. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. In addition, the sensitivity of the radiographic technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25, as determined in accordance with 4.5.2.2. In the event an original radiograph leaves doubt as to acceptability, the fuze shall be re-radiographed at an angle 90° from that of the original exposure and in the same plane.

4.5.13.1 Sensitivity check of radiographic technique. The sensitivity of radiograhic technique shall be determined by the means of a penetrameter fabricated as specified herein.

Location of the penetrameter hole shall be TBD inches measured from base shoulder of fuze body towards fuze nose. The fuze, with attached penetrameter, shall be positioned between the radiograhic source and the radiographic film with the penetrameter facing the radiographic source. When radiographing a group of fuzes on a rotating ring, the penetrameter, taped to a fuze, may be used to represent the group when checking sensitivity. The sensitivity of radiographic technique shall be considered satisfactory when the image of the penetrameter, including the holes, is sharply defined on the radiograph. The penetrameters shall be made of steel having an approximate specific gravity of 7.8. Penetrameter thickness shall be .020 plus or minus (±) .010 inches. Each penetrameter shall have a circular hole with a diameter twice that of the penetrameter thickness. For further details on construction of the penetrameters, see MIL-STD-453. Frequency of sensitivity check shall be no less than one in each fifty (50) radiographs.

4.5.13.2 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of the maximum equivalent thickness to be radiographed. Film density shall be checked each hour of operation, or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. To judge densities, it is recommended that standards be used. Comparison of the film with standards may be made visually; however, the use of a densitometer is recommended.

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5. PACKAGING

5.1 Packaging requirements. The fuze parts shall be packaged in accordance with the requirements of MIL-STD-1169 and dwg. 8865235.

5.2 Packing and marking. The fuze parts shall be packed and marked in accordance with the requirements of MIL-STD-1169 and dwgs. 8865236 and 8865235.

6. NOTES

6.1 Intended use. The parts and assemblies covered by this specification are intended for use on the XM768 fuze, formerly designated the M84A1E1 fuze. This fuze is intended for use on the Cartridge, 81MM, Illum, XM853.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of designs for approval. See MIL-A-48078. Submit inspection equipment designs as required to; Commander, US Army ARDC, ATTN: AMSMC-QAT-I(D), Dover, NJ 07801-5001.

6.4 Proving ground test summary.

6.4.1 First article test and first three production lots.

Cartridge	Charge	Set time	Quantity
XM853	1	5	25
	2	15	25
	3	30	25
	4	55	25

6.4.2 Production.

Cartridge	Charge	Set time	Quantity
XM853	2	15	25
	3	30	25

6.5 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be

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delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.	Option
4.5.6	Test & Demonstration Reports	DI-T-1906	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.6 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal ARRADCOM or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

6.7 Standard deviation. The following unbiased formula for standard deviation shall be utilized:

$$\text{standard deviation} = \sqrt{\frac{(Xi - X)^2}{n-1}}$$

$$(n-1)$$

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(Project 1390-A499)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-70601	Fuze, Time, XM768 Metal Parts LAP

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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